UNIT 1 · 8 hours

# Block Diagram Algebra and Reduction Techniques



Introduction and Mathematical Modelling of Physical Systems

**Control Systems (22EEG53)**
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# Learning objectives

- Identify block, summing point and take-off point
- Apply series, parallel and feedback reduction rules
- Shift summing and take-off points
- Reduce a multi-loop block diagram to a single transfer function

**Syllabus context**

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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# Elements of a block diagram

**Block:** rectangle containing the transfer function; output = G × input.

**Summing point:** circle with ± signs; algebraic sum of signals.

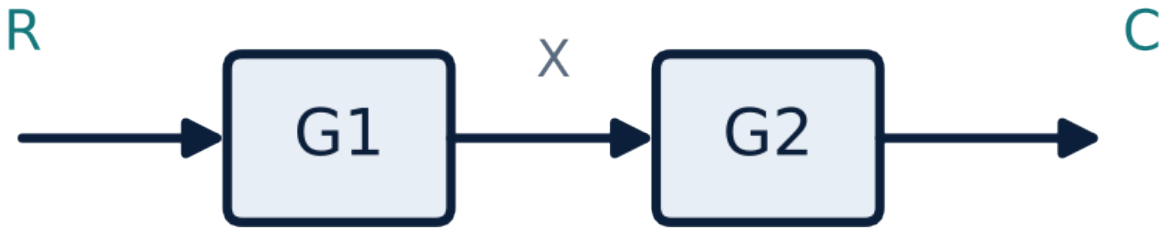
**Take-off point:** dot: the same signal is branched off without loading.

**Arrows:** direction of signal flow.

**Why reduce?** to get one overall transfer function C(s)/R(s).

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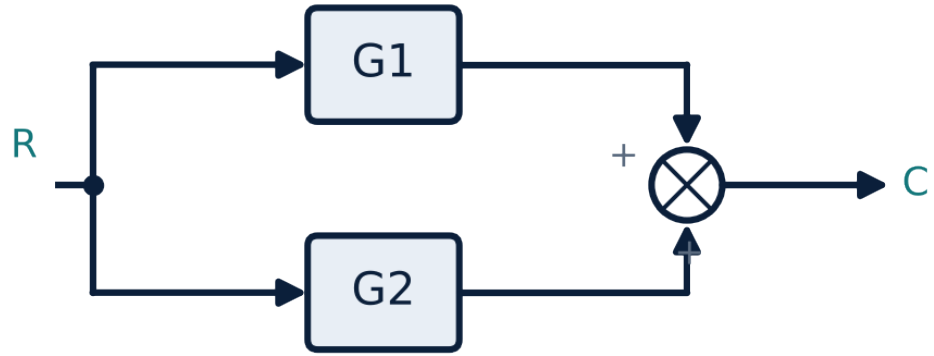
# Rule 1: blocks in series (cascade)




- C/R = G1 · G2
- Valid when blocks do not load each other.
- Any number of blocks: multiply all.

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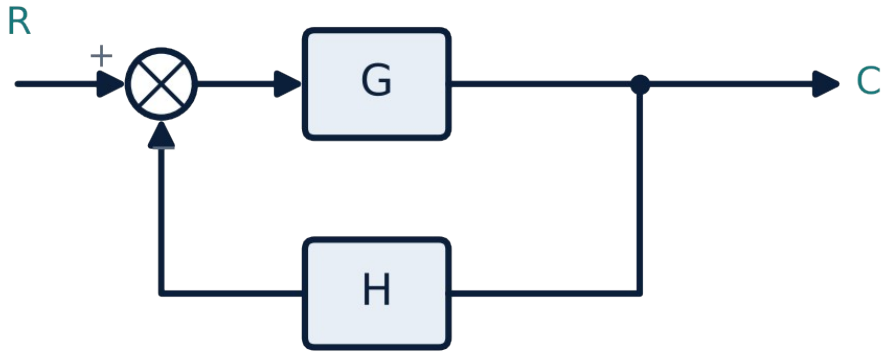
# Rule 2: blocks in parallel




- C/R = G1 ± G2
- Signs follow the summing point.
- Same input to both blocks.

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# Rule 3: feedback loop




- C/R = G / (1 ± GH)
- Negative feedback: 1 + GH; positive: 1 − GH.
- Unity feedback: H = 1 → G/(1 + G).
- GH = open-loop transfer function.

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# Moving summing and take-off points

| Operation | Rule |
|---|---|
| **Move take-off point beyond a block G** | Add a block G in the branch |
| **Move take-off point before a block G** | Add a block 1/G in the branch |
| **Move summing point beyond a block G** | Add G in the branch being moved |
| **Move summing point before a block G** | Add 1/G in the branch being moved |
| **Interchange adjacent summing points** | Allowed |
| **Interchange summing and take-off points** | Not allowed (shift across a block) |

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# Reduction procedure

1

**Combine series**

Multiply cascade blocks.

2

**Combine parallel**

Add/subtract blocks with common input and output.

3

**Eliminate inner loops**

Apply $G/(1 \pm GH)$ to the innermost feedback.

4

**Shift points**

Only when needed to untangle.

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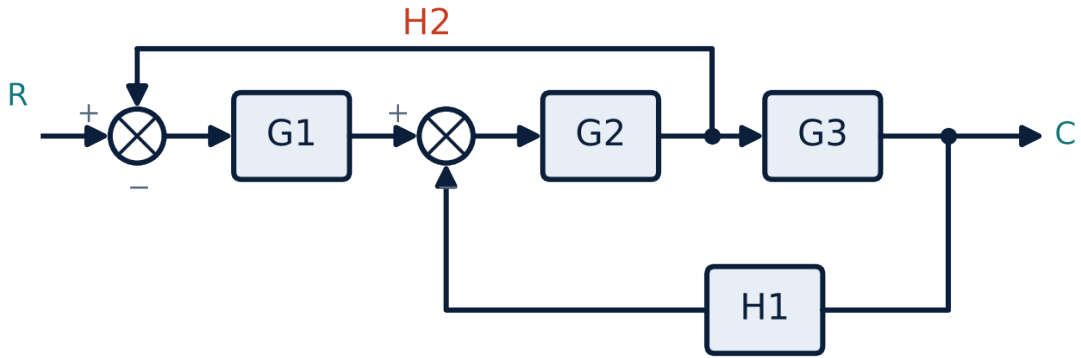
**Repeat**

Until a single block remains.

**Rule of thumb: reduce inner loops first and keep the same input/output.**

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# Example: multi-loop diagram




- Inner loop: G2 G3 with feedback H1.
- Then outer loop: G1 with feedback H2 taken after G2.
- Reduce step by step.

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# Reducing the example

**Problem** G1 = 10, G2 = 2, G3 = 5, H1 = 0.1, H2 = 0.2. Find C/R.

- Inner loop (G2G3 with H1): G2G3/(1 + G2G3H1) = 10/(1 + 1) = 5
- Path u1 → G2 output: G2/(1 + G2G3H1) = 2/2 = 1
- Overall: C/R = G1G2G3 / (1 + G2G3H1 + G1G2H2)
- = 100 / (1 + 1 + 4)

**Answer: C/R = 100/6 = 16.67**

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# Key takeaways

- Series: G1G2;  parallel: G1 ± G2;  feedback: G/(1 ± GH)
- Move take-off points and summing points to expose simple forms
- Reduce inner loops first
- Check signs at every summing point

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# Practice questions

**Q1.** Reduce a unity-feedback system with G = 10/(s(s+2)) to a single block.

**Q2.** State the rules for shifting a take-off point across a block.

**Q3.** G1 = 1/(s+1), G2 = 1/(s+2), H = 1: find C/R.

**Q4.** Why can't a summing point and a take-off point be interchanged directly?